

Modulation

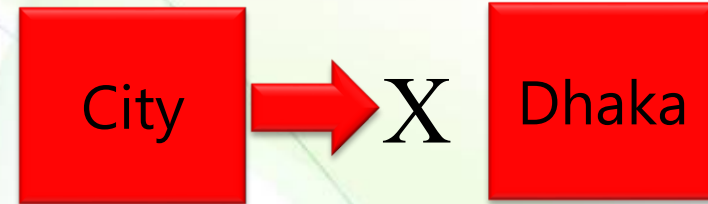
Need For Using High Carrier Frequency

- ❖ **Radiation efficiency** (Increase/ decrease?)
- ❖ **Size of antenna** (Large/ small?)
- ❖ **Range of Communication** (Short/ long?)
- ❖ **Easy of selection of radio signal in the receiver** (Why?)

Modulation (1/2)



Only a Person Can go



Only a Person Cannot go



Modulation

Information

Carrier



Person
using
bus
Can go

Modulation (2/2)

❖ Definition:

It is a process of vary some characteristics of the Carrier Voltage in accordance with the Modulating Voltage

- Amplitude
- Frequency
- Phase

**Information Signal
(What?)**

High Frequency Signal,
$$v_c = V_c \cos(\omega_c t + \phi)$$

**Phase of the
carrier**

Freq. of the carrier

Amplitude of the carrier

Applications & Types of Modulation

● Application Examples

- Broadcasting of both audio and video signals.
- Mobile radio communications, such as cell phone.

□ Types of Modulation:

❖ Amplitude Modulation

- Amplitude of carrier is varied with respect to modulating signal

❖ Frequency Modulation

- Frequency of carrier is varied with respect to modulating signal

❖ Phase Modulation

- Phase angle of carrier is varied with respect to modulating signal

Need for Modulation

- 1) The modulating signal produced by microphone or camera is very weak and very low frequency (Why?)

We know energy in a signal is proportional to its frequency

- So this signal can not travel long distance. That is why, weak signal requires high frequency carrier.

2) Height of antenna needed

For best result antenna height = the wavelength

we know $v = f\lambda$

Example: if signal is 20KHz

Then antenna height = $v/f = 15000$ m (**Practical?**)

V = velocity of light (3×10^8)

f = frequency

λ = wavelength

- 3) Modulation permits the transmission without wire.

Need for Modulation

- 4. Modulation makes the message signal more robust against the noise.**
- 5. Modulation is required to match the signal frequency with the bandwidth of the channel.**

Complete Classification of Modulation Processes (1/2)

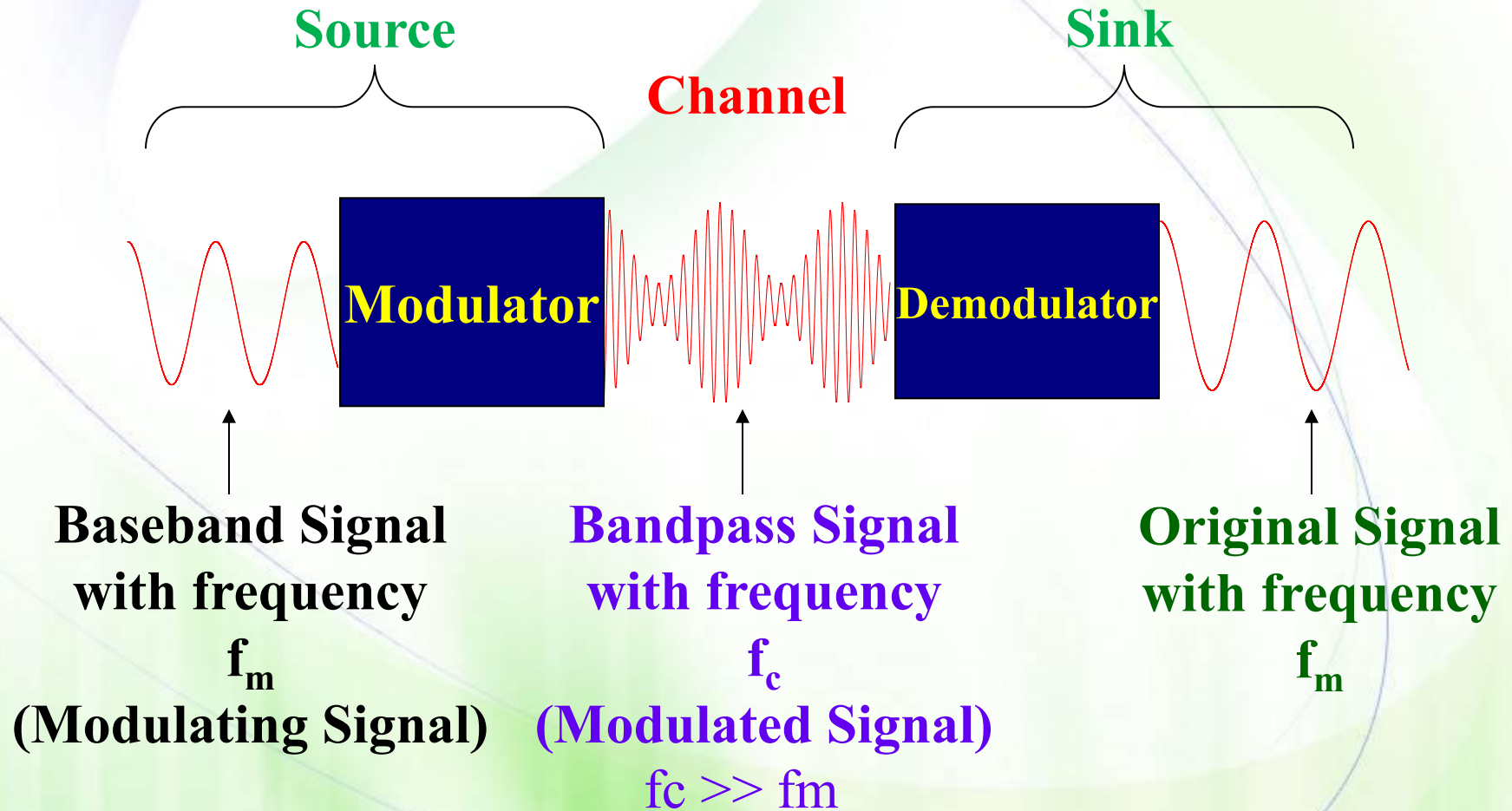
- **Amplitude Modulation (AM)**
 - i) **Single sideband AM (SSBAM)**
 - ii) **Double sideband AM (DSBAM)**
 - iii) **Frequency division multiplexing (FDM)**
- **Frequency Modulation (FM)**
- **Phase Modulation**
- **Pulse Modulation**
 - **Pulse amplitude modulation (PAM)**
 - **Time Division multiplexing (TDM)**
 - **Pulse time modulation (PTM)**
 - **Pulse division multiplexing**
 - **Pulse code modulation (PCM)**

Complete Classification of Modulation Processes (2/2)

- **Digital modulation (DM)**
 - **Differential PCM (DPCM)**
 - **Adaptive PCM (ADPCM)**
 - **Delta modulation (DM)**
 - **Adaptive delta modulation (ADM)**
- **Multiplexing**

Amplitude Modulation (AM)

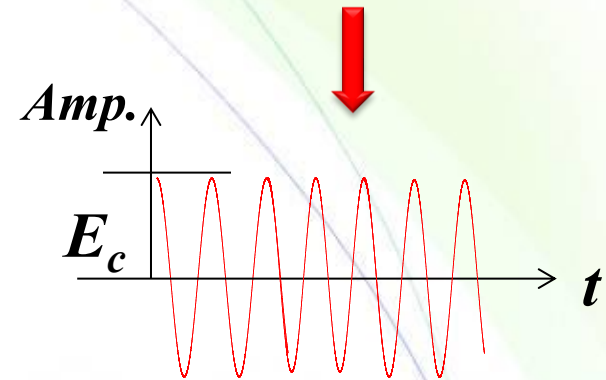
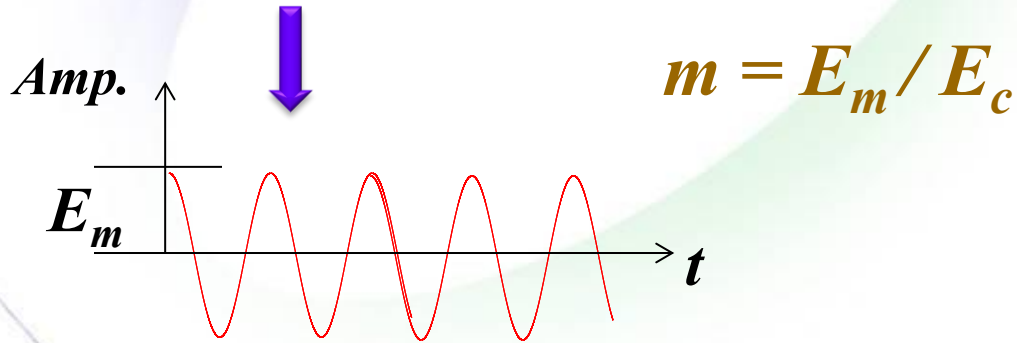
- ❑ **Amplitude** of the **carrier** is varied according to **amplitude** of the **modulating signal**



Voice: 300-3400Hz

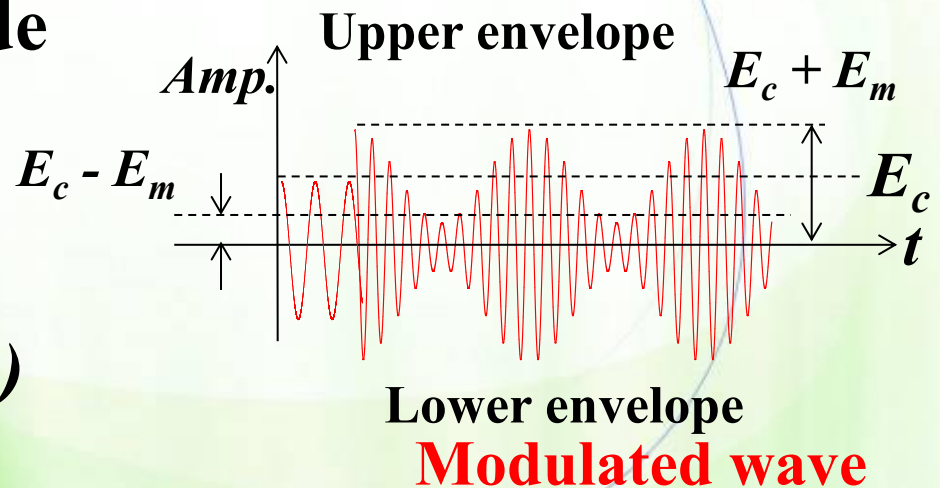
Modulation Index or Factor (m) (1/3)

(1) it is the ratio of maximum value of the modulating signal to the maximum amplitude of the carrier



(2) It is the ratio of change of amplitude of the carrier to its original amplitude

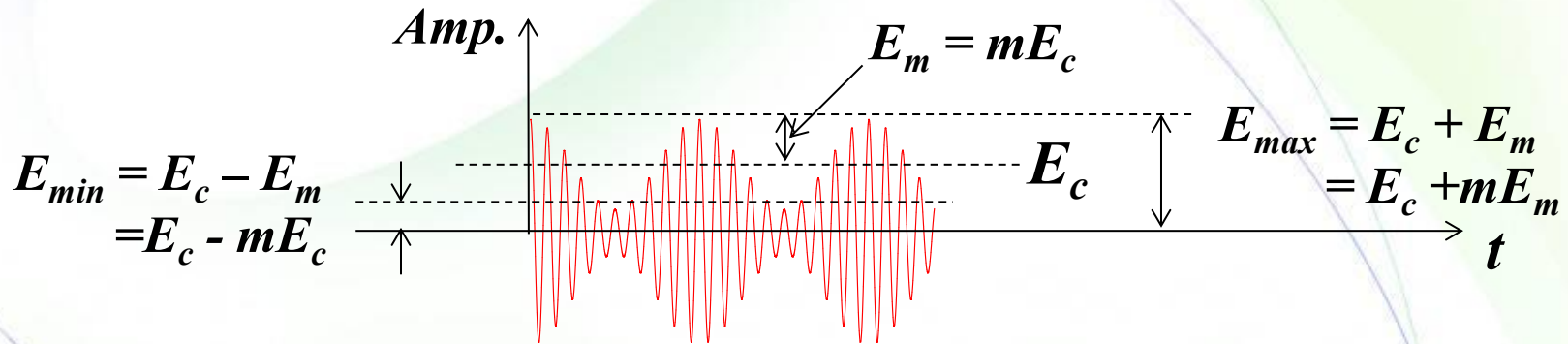
$$m = \Delta E_c / E_c$$



(3) $m = 100 * (\Delta E_c / E_c)$

Modulation Index or Factor (m) (2/3)

(4) It is the ratio of **minimum amplitude** to the **maximum amplitude** of the modulated wave



$$m = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}}$$

Proof: $E_{\max} = E_c + mE_c$ (1)

$$E_{\min} = E_c - mE_c$$
 (2)

Modulation Index or Factor (m) (3/3)

Adding Equations (1) and (2) we have:

$$E_c = \frac{E_{\max} + E_{\min}}{2}$$

Subtracting Equations (1) and (2) we have:

$$E_m = \frac{E_{\max} - E_{\min}}{2}$$

Now:

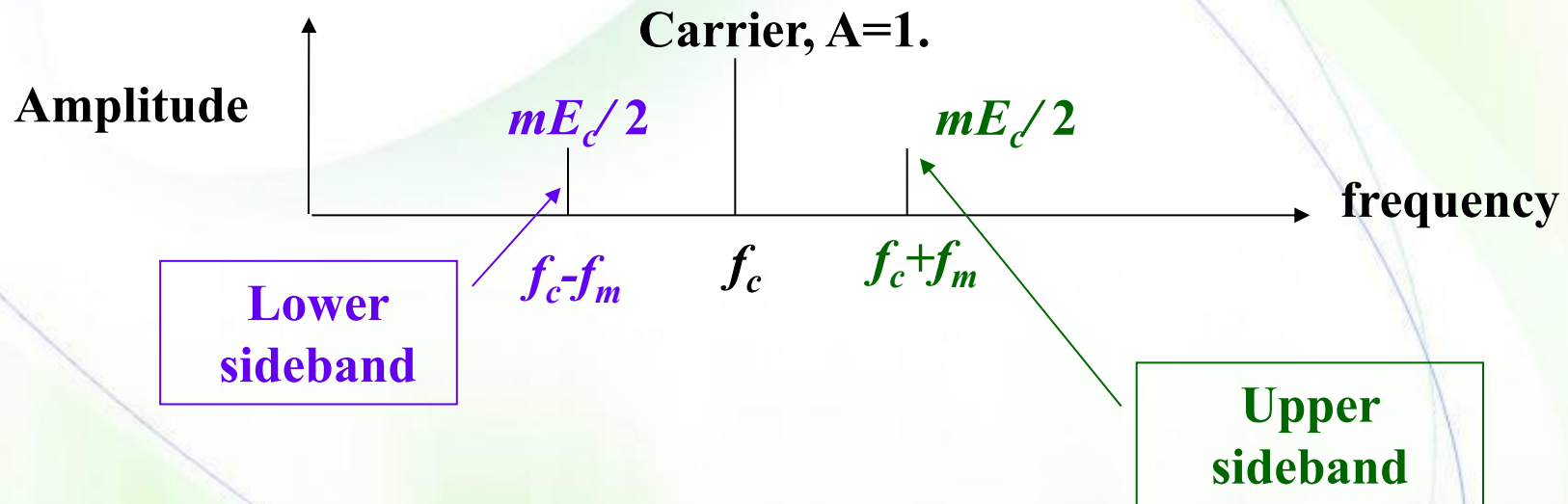
$$m = \frac{E_m}{E_c} = \frac{E_{\max} - E_{\min}}{E_{\max} + E_{\min}}$$

Note: 1) The value of m lies between 0 & 1

2) It is also depends on Amplitude of modulating signal and carrier

Mathematical Analysis of Amplitude Modulated Wave (3/3)

Frequency spectrum of the above modulated wave:

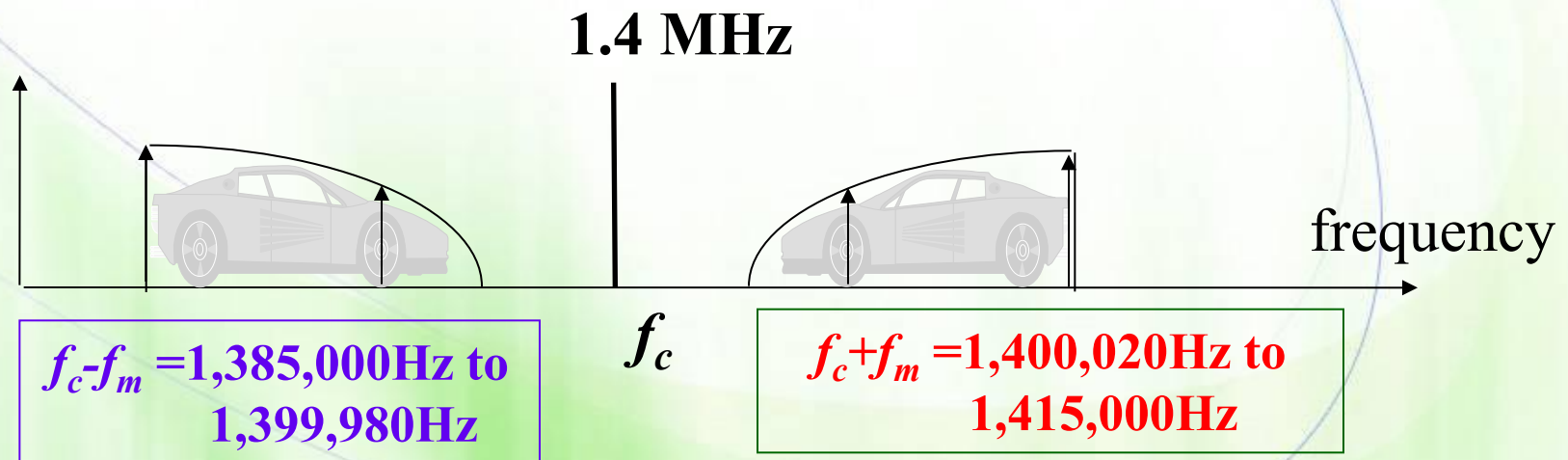


$$\begin{aligned}\text{Bandwidth} &= \text{USB} - \text{LSB} \\ &= (f_c + f_m) - (f_c - f_m) \\ &= 2f_m\end{aligned}$$

Example of Amplitude Modulation

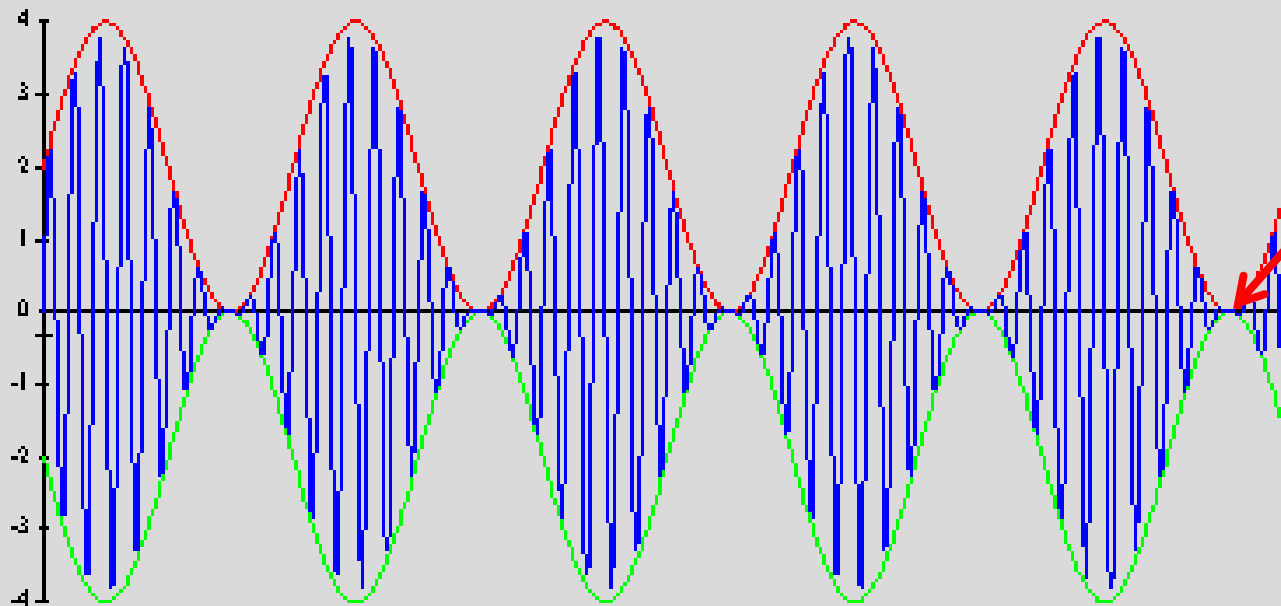
- The information signal is usually not a single frequency but a range of frequencies (i.e., band). For example, frequencies from **20Hz to 15KHz**. If we use a carrier of **1.4MHz**, what will be the AM spectrum?

Solⁿ: In frequency domain the AM waveform are the **lower-side band frequency ($f_c - f_m$)**, the **carrier frequency f_c** , the **upper-side frequency/band ($f_c + f_m$)**. Bandwidth: $2 \times (15\text{K} - 20)\text{Hz}$.



Modulation Index of AM (1/3)

Modulation Index = 1

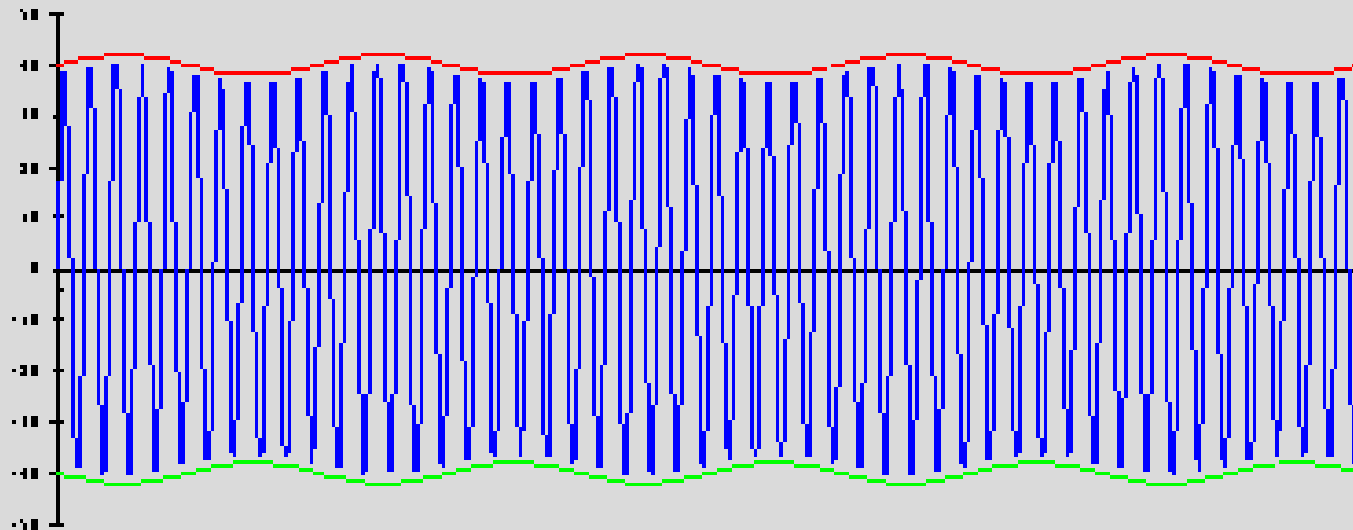


**100%
Modulation**

Modulation Index of AM (2/3)

Modulation Index =.05

Max. Amp. = 2v, DC of 40v added

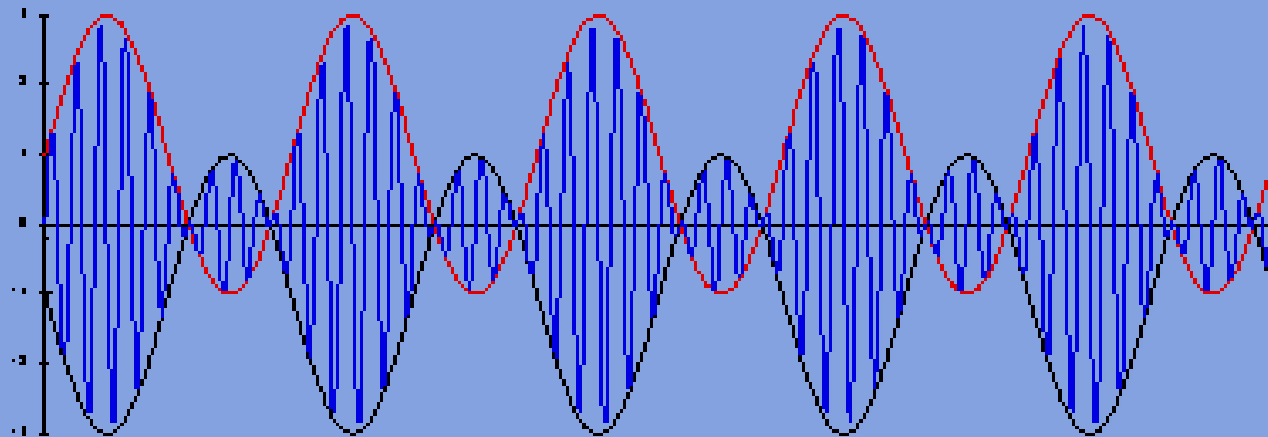


Undermodulation

Modulation Index of AM (3/3)

Modulation Index = 2

Max. Amp. = 2v, DC of 1v added



Overmodulation

Power Distribution in the AM Wave (1/4)

□ We know power in a voltage wave $\propto (\text{amplitude})^2$

○ **Total power in an AM wave, $P_T = P_{(LSB)} + P_C + P_{(USB)}$**

○ **Power contained in carrier is:**

$$P_C = \left(\frac{E_c}{\sqrt{2}} \right)^2 = \frac{E_c^2}{2} \quad \left[\begin{array}{l} E_c = \text{Max}^m \text{ value} \\ \frac{E_c}{\sqrt{2}} = \text{R.M.S. value} \end{array} \right]$$

○ **Power contained in LSB is:**

$$P_{(LSB)} = \left(\frac{1}{\sqrt{2}} \frac{mE_c}{2} \right)^2 = \frac{m^2 E_c^2}{8} = \frac{m^2}{4} \times \frac{E_c^2}{2} = \frac{P_C m^2}{4}$$

○ **Power contained in USB is:**

$$P_{(USB)} = \left(\frac{1}{\sqrt{2}} \frac{mE_c}{2} \right)^2 = \frac{m^2 E_c^2}{8} = \frac{m^2}{4} \times \frac{E_c^2}{2} = \frac{P_C m^2}{4}$$

Power Distribution in the AM Wave (2/4)

➤ **Total power in AM wave is:**

$$P_T = \frac{P_C m^2}{4} + P_C + \frac{P_C m^2}{4}$$

$$P_T = P_C + \frac{P_C m^2}{2}$$

$$P_T = P_C \left(1 + \frac{m^2}{2}\right)$$

$$\frac{P_T}{P_C} = \left(1 + \frac{m^2}{2}\right)$$

Power Distribution in the AM Wave (3/4)

➤ **Power in both side bands is:**

$$P_{SB} = P_{LSB} + P_{USB}$$

$$P_{SB} = \frac{P_C m^2}{4} + \frac{P_C m^2}{4} = \frac{P_C m^2}{2}$$

➤ **Ratio of sideband power to total power is:**

$$\frac{P_{SB}}{P_T} = \frac{P_C m^2 / 2}{P_C (1 + \frac{m^2}{2})}$$

$$\frac{P_{SB}}{P_T} = \frac{m^2}{2 + m^2}$$

Power Distribution in the AM Wave (4/4)

- For 100% modulation $m = 1$

$$P_{SB} = \frac{1}{3} P_T$$

- So the side bands contain 1/3rd (33%) of the total power of the AM wave

Example

1) A transmitter supplies 10Kw power to an aerial when unmodulated. Determines the power radiated when modulated to 30%.

Sol: Here, $P_c = 10\text{Kw}$

$$m = 0.3$$

$$P_T = ?$$

$$\begin{aligned}\text{We know, } P_T &= P_c(1+m^2/2) \\ &= 10[1+(0.3)^2/2] \\ &= 10.45 \text{ Kw}\end{aligned}$$

Example

2) A 100v, 10KHz carrier is modulated with the help of a 5v, 50Hz signal. Calculate-

- a. modulation factor(m)
- b. Amplitude of each sideband
- c. Frequency of each sideband
- d. Bandwidth of modulated wave

Sol: (a) $m = \frac{E_m}{E_c} = \frac{5}{100} = 0.05$

(b) Amplitude of each sideband $= \frac{mE_c}{2} = \frac{0.05 * 100}{2} = 2.5v$

(c) Frequency of sidebands, USB $= f_c + f_m = 10000 + 50 = 10050\text{Hz}$
LSB $= f_c - f_m = 10000 - 50 = 9950\text{Hz}$

(d) Bandwidth BW = USB - LSB = 10050 - 9950 = 100Hz

Example

3) A 100v, 100KHz carrier is modulated with the help of a 10v, 1KHz signal to the extent of 50%. Write down the equation for the AM wave.

Sol: Here, $m=0.5$, $E_c = 100\text{v}$, $E_m = 10\text{v}$, $f_c = 100\text{KHz}$, $f_m = 1\text{KHz}$

We know AM wave is:

$$e = E_c \sin \omega_c t + \frac{mE_c}{2} \cos(\omega_c - \omega_m)t - \frac{mE_c}{2} \cos(\omega_c + \omega_m)t$$

$$e = E_c \sin(2\pi f_c)t + \frac{mE_c}{2} \cos(2\pi f_c - 2\pi f_m)t - \frac{mE_c}{2} \cos(2\pi f_c + 2\pi f_m)t$$

Example

$$e = 100\sin 628000t + \frac{0.5 * 100}{2} \cos(628000 - 6280)t - \frac{0.5 * 100}{2} \cos(628000 + 6280)t$$

$$e = 100\sin 628000t + 25\cos 621720t - 25\cos 634280t$$

Example

4) An AM wave is represented by the equation.

$$e = 20(1 + 0.7 \sin 6280t) \sin 628000t$$

Determine (1) m , (2) E_c , (3) f_m , (4) f_c , (5) E_{max} , (6) E_{min} , (7) BW

Sol: We know standard AM wave equation

$$e = E_c (1 + m \sin \omega_m t) \times \sin \omega_c t$$

Comparing this equation with the above equation. We get

(1) $m = 0.7$

(2) $E_c = 20$

(3) $f_m = 1\text{KHz}$ (How?)

(4) $f_c = 100\text{KHz}$ (How?)

(5) $E_{max} = E_c + m E_c = 34\text{v}$

(6) $E_{min} = E_c - m E_c = 6\text{v}$

(7) $BW = 2f_m = 2\text{KHz}$

Example

5) Determine the percentage of the ratio of the sidebands power to the total output power of the AM wave for $m = 0.5$ and $m = 0.3$.

Solⁿ: for $m=0.5$

$$\frac{P_{SB}}{P_T} = \frac{m^2}{2 + m^2} \times 100\% = \frac{(0.5)^2}{2 + (0.5)^2} \times 100\% = 11.11\%$$

$m = 0.3$

$$\frac{P_{SB}}{P_T} = \frac{m^2}{2 + m^2} \times 100\% = \frac{(0.3)^2}{2 + (0.3)^2} \times 100\% = 4.3\%$$

Limitations of The AM

- The useful power is contained in the sidebands & even at 100% modulation the sidebands contain only 33% of the total power & hence the modulation efficiency is poor.
- Due to poor efficiency transmitter has very poor range.
- The reception in this modulation is noisy.
- The audio reception is poor quality due to noisy reception.

Introduction

- ❑ In AM, the amplitude of a carrier is modulated by a signal $m(t)$.
- ❑ Other two parameters of the carrier, i.e., phase and frequency, can also be varied then these modulations are called:
 - Frequency modulation (Define?)
 - Phase modulation (Define?)

Frequency Modulation (FM) (1/3)

- Frequency f_c of a carrier signal changes with the modulating signal:
- The bandwidth of FM is high. It is approximately = $10 \times$ signal frequency.
- The total bandwidth required for FM is:

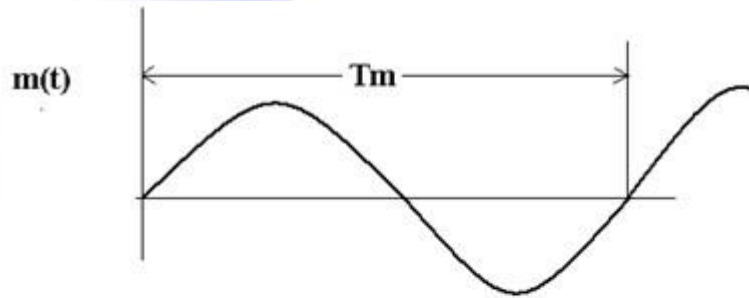
$$B_{FM} = 2(1 + \beta)B$$

Here, usual value of β is 4.

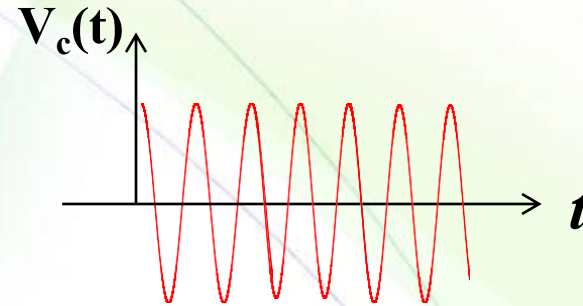
Frequency Modulation (FM) (1/3)

- Generation of FM modulator required:
 - A voltage-to-frequency converter VFC
 - A voltage controlled oscillator VCO

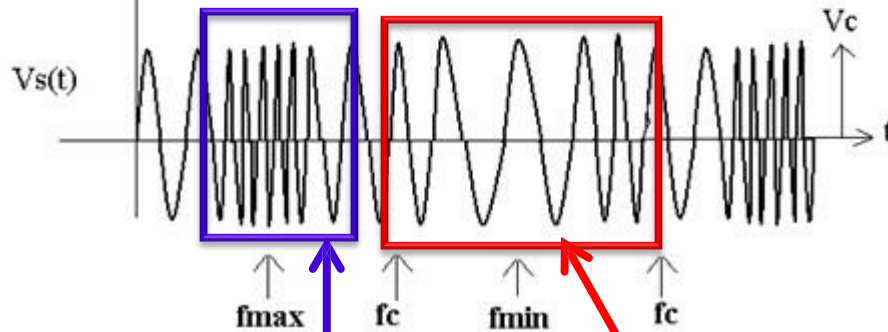
Frequency Modulation (FM) (2/3)



Modulating signal



Carrier signal



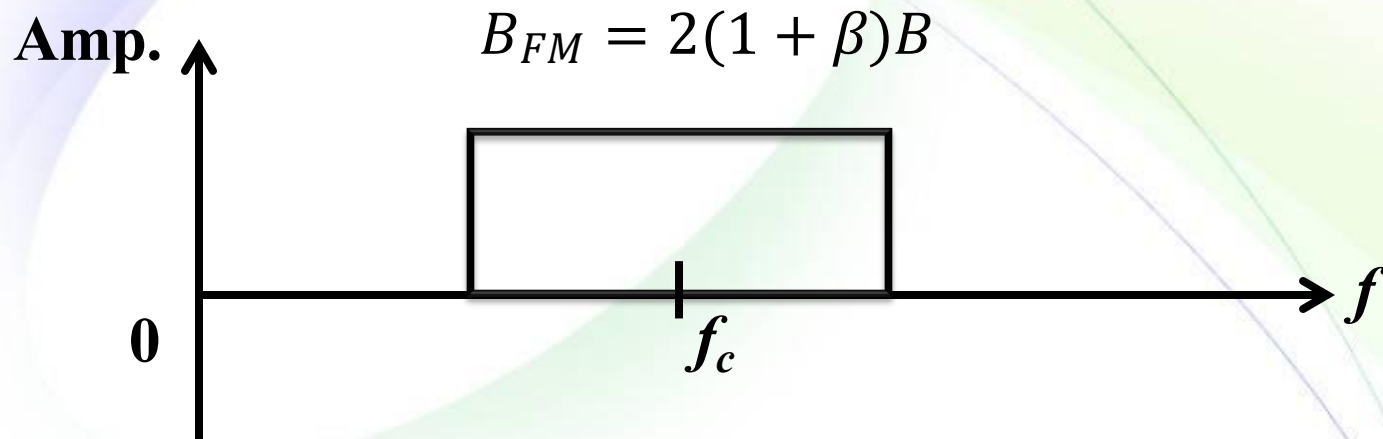
FM signal

Frequency Modulation (FM) (2/3)

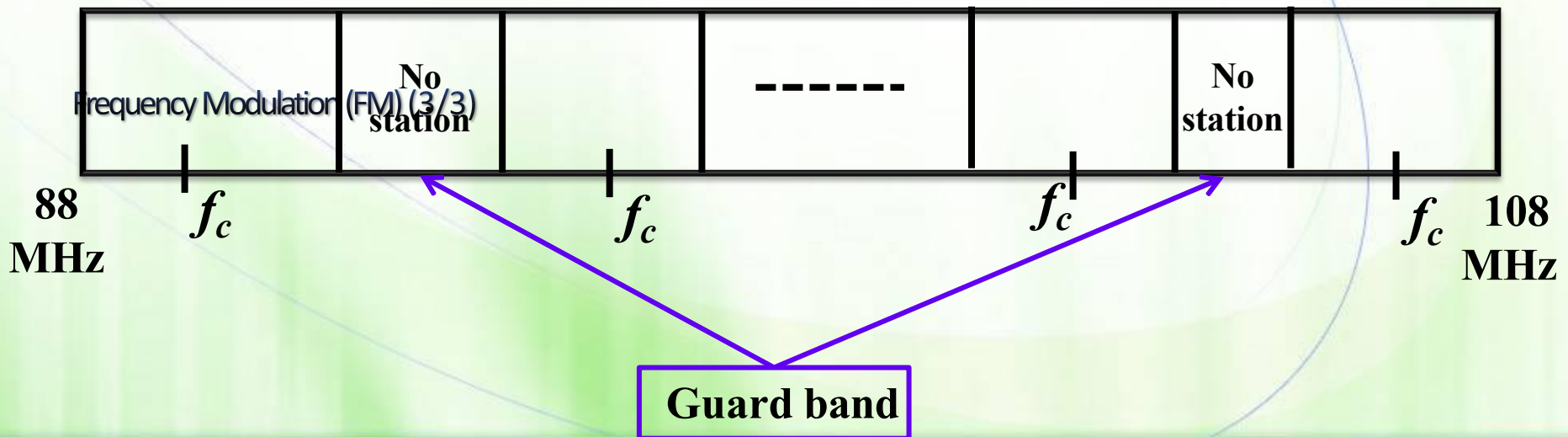
**Signal Amplitude low
frequency is also low**

**Signal Amplitude high
frequency is also high**

Frequency Modulation (FM) (3/3)



FM Band Allocation:



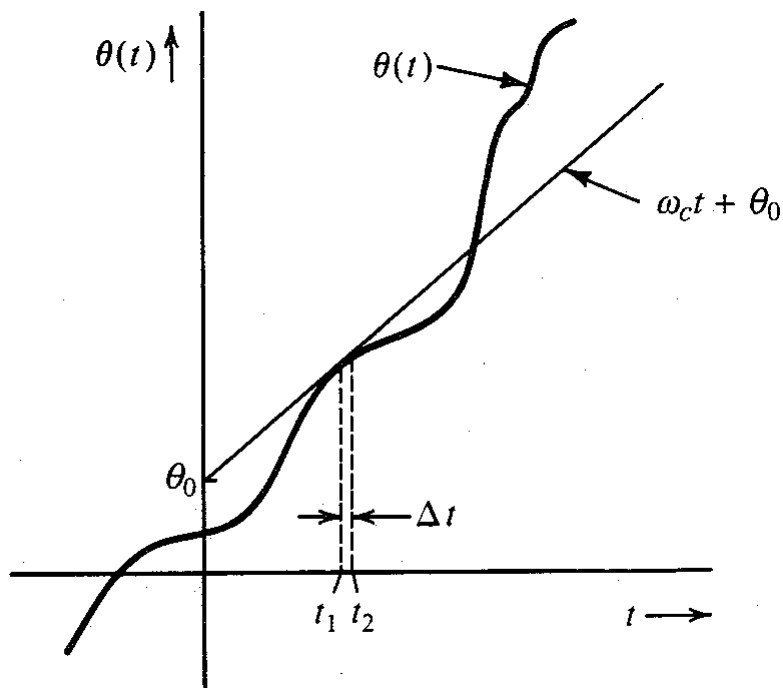
Concept of Instantaneous Frequency (1/4)

- ❖ AM signals carry a message with their varying amplitude
- ❖ FM signals vary the instantaneous frequency in proportion to the modulating signal $m(t)$
- ❖ This means that the carrier frequency is changing continuously at every instant.
- ❖ Let us consider a generalized sinusoidal signal $\phi(t)$ given by

$$\phi(t) = A \cos \theta(t)$$

here, $\theta(t)$ is the generalized angle and is a function of t .

Concept of Instantaneous Frequency (2/4)



Here, generalized angle for a conventional sinusoid $A \cos(\omega_c t + \theta_0)$ is $(\omega_c t + \theta_0)$. Over a small interval $\Delta t \rightarrow 0$ the signal $\phi(t) = A \cos \theta(t)$ and the sinusoid $A \cos(\omega_c t + \theta_0)$ are identical. That is -

$$\phi(t) = A \cos(\omega_c t + \theta_0)_{t_1 < t < t_2}$$

So at small interval Δt , the angular frequency of $\phi(t)$ is ω_c . Because $(\omega_c t + \theta_0)$ is tangential to $\theta(t)$, the angular frequency of $\phi(t)$ is the slope of its angle $\theta(t)$ over the small interval. So at every instant the instantaneous frequency ω_i at any instant t is the slope of $\theta(t)$.

Concept of Instantaneous Frequency (3/4)

$$\omega_i(t) = \frac{d\theta}{dt}$$

$$\theta(t) = \int_{-\infty}^t \omega_i(\alpha) d\alpha$$

Again the angle of a carrier is varied with a modulating signal $m(t)$ are known as angle modulation. In phase modulation (PM), $\theta(t)$ is varied linearly with $m(t)$.

$$\theta(t) = \omega_c t + \theta_0 + k_p m(t)$$

where, k_p is a constant and ω_c is the carrier frequency. Assume

$$\theta(t) = \omega_c t + k_p m(t)$$

The PM wave is –

$$\varphi_{\text{PM}}(t) = A \cos [\omega_c t + k_p m(t)]$$

Concept of Instantaneous Frequency (4/4)

In this case, the instantaneous angular frequency $\omega_i(t)$ is

$$\omega_i(t) = \frac{d\theta}{dt} = \omega_c + k_p \dot{m}(t)$$

where, $\dot{m}(t)$ is the derivative of $m(t)$

So in PM, the instantaneous angular frequency $\omega_i(t)$ varies linearly with the derivative of the modulating signal.

In FM, the instantaneous angular frequency $\omega_i(t)$ is

$$\omega_i(t) = \omega_c + k_f m(t)$$

where, k_f is a constant and the angle $\theta(t)$ is -

$$\theta(t) = \int_{-\infty}^t [\omega_c + k_f m(\alpha)] d\alpha$$

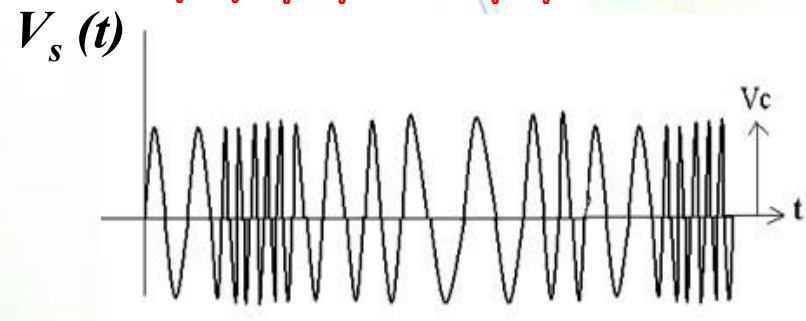
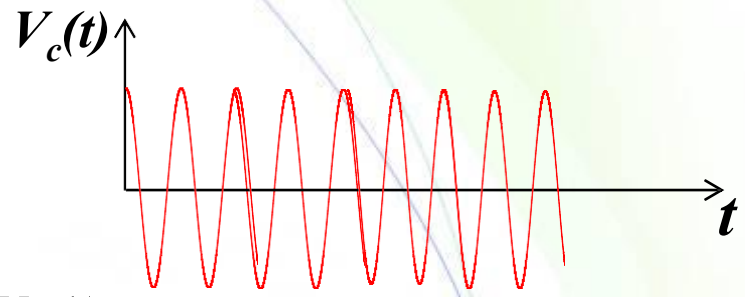
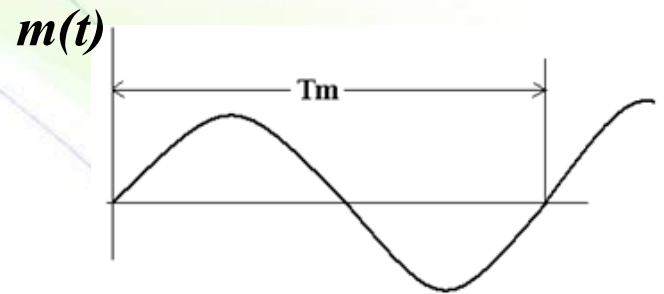
$$= \omega_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha$$

$$\varphi_{\text{FM}}(t) = A \cos \left[\omega_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha \right]$$

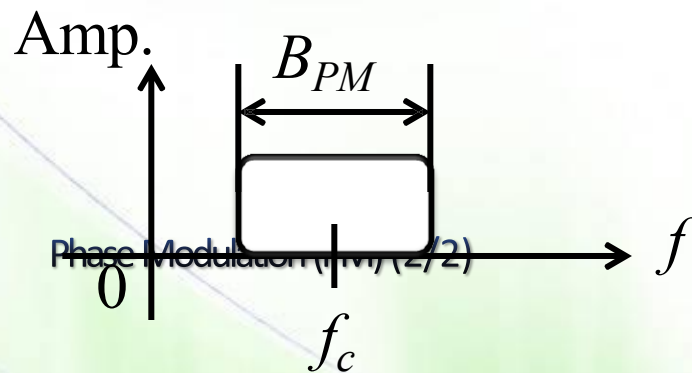
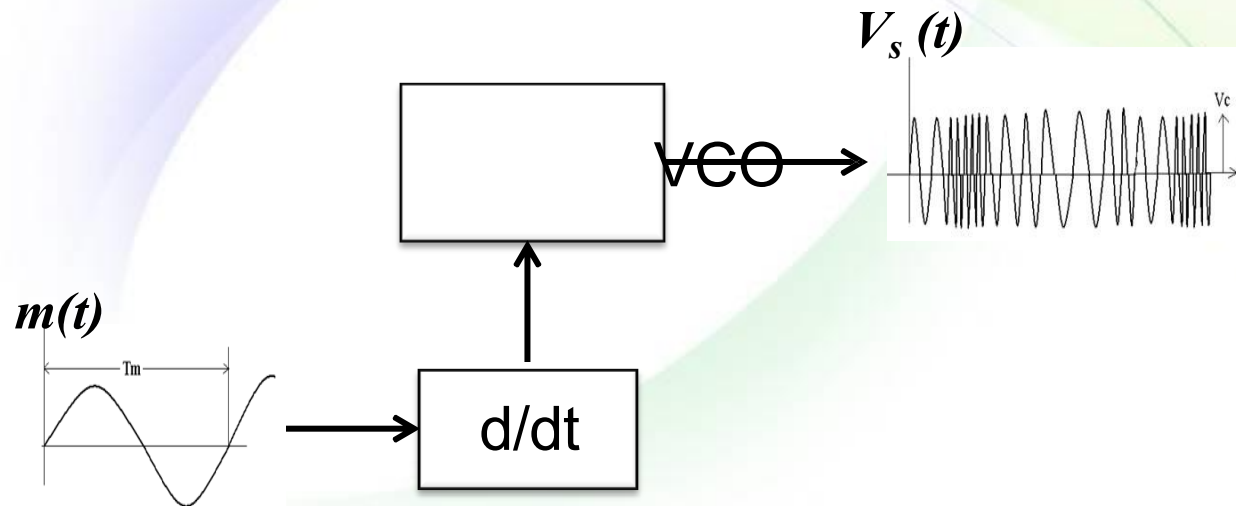
❖ The modulating signal only changes the phase of the carrier signal

❖ The phase changes manifests itself as a frequency change but the instantaneous frequency change is proportional to the derivative of the amplitude

❖ The bandwidth is higher than for AM



Phase Modulation (PM) (1/2)



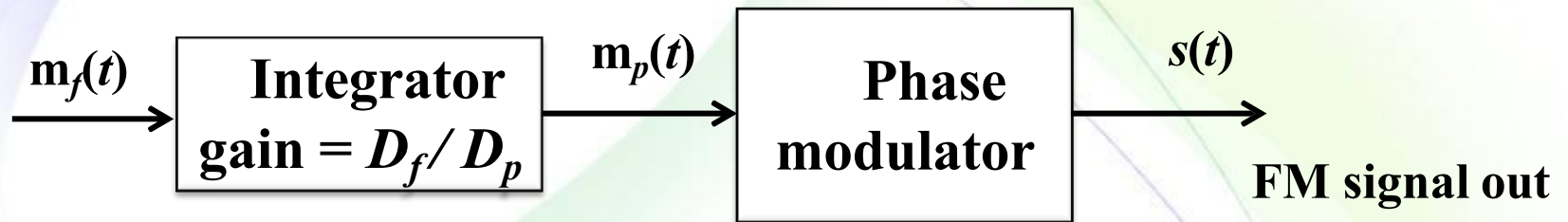
**Total bandwidth required
for PM –**

$$B_{PM} = 2(1 + \beta)B$$

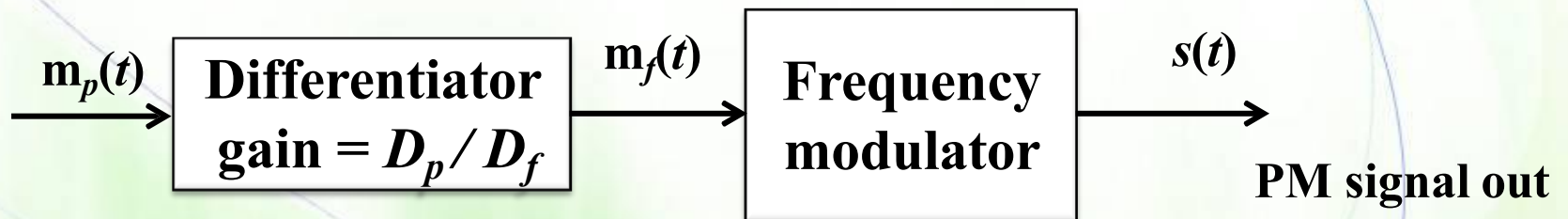
where, $\beta = 2$

Generation of FM from PM and Vice Versa

Generation of FM using Phase modulator



Generation of PM using Frequency modulator



Generation of FM from PM and Vice Versa

- **FM:**

$$m_f(t) = \frac{D_p}{D_f} \left[\frac{dm_p(t)}{dt} \right]$$

**D_p = gain of PM
ckt.**

**D_f = gain of FM
ckt.**

- **PM:**
$$m_p(t) = \frac{D_f}{D_p} \int_{-\alpha}^t m_f(\alpha) d\alpha$$

If a bandpass signal is represented by

$$s(t) = R(t) \cos \phi(t)$$

where, $\phi(t) = \omega_c(t) + \theta(t)$

The instantaneous frequency of $s(t)$ is –

$$f_i(t) = \frac{1}{2\pi} \omega_i(t) = \frac{1}{2\pi} \left[\frac{d\phi(t)}{dt} \right]$$

or

$$f_i(t) = f_c + \left[\frac{d\theta(t)}{2\pi dt} \right]$$

Generation of FM from PM and Vice Versa

- **For the case of FM we obtain the instantaneous frequency**

$$f_i(t) = f_c + \frac{1}{2\pi} D_f m(t)$$

- **The above equation represents FM because the instantaneous frequency varies about the assigned carrier frequency f_c which is directly proportional to the $m(t)$**

- **The frequency deviation from the carrier frequency is –**

$$f_d = f_i(t) - f_c = \frac{1}{2\pi} \left[\frac{d\theta(t)}{dt} \right]$$

- **The peak frequency deviation is related to the peak modulating voltage by –**

$$\Delta F = \frac{1}{2\pi} D_f V_p$$

where,
 $V_p = \max[m(t)]$

Generation of FM from PM and Vice Versa

- Similarly, the peak phase deviation is -

$$\Delta\theta = D_p V_p$$

- The frequency modulation index is –

$$\frac{\Delta F}{B} \quad \beta_f =$$

where, B is the bandwidth of modulating signal

- The phase modulation index is -

$$\beta_p = \Delta\theta$$